

☰ VPC > Your VPCs > Create VPC ⓘ

Create VPC ⓘ

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings ⓘ

Resources to create ⓘ
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional ⓘ
Creates a tag with a key of 'Name' and a value that you specify.

new-vpc

IPv4 CIDR block ⓘ
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block ⓘ
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tags ⓘ

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Q Name X

Value - optional
Q new-vpc X Remove tag

Add tag
You can add 49 more tags

Cancel Preview code Create VPC

Step 5 the following dashboard will appear

aws ☰ Search [Alt+S] Asia Pacific (Mumbai) nikita

☰ VPC > Your VPCs > vpc-0df59c88041a47a21 ⓘ

VPC dashboard

Filter by VPC

AWS Global View ⓘ

- Virtual private cloud
 - Your VPCs**
 - Subnets
 - Route tables
 - Internet gateways
 - Egress-only internet gateways
 - DHCP option sets
 - Elastic IPs
 - Managed prefix lists
 - NAT gateways
 - Peering connections
 - Route servers
- Security
 - Network ACLs
 - Security groups

You successfully created vpc-0df59c88041a47a21 / new-vpc

vpc-0df59c88041a47a21 / new-vpc Actions

Details ⓘ

VPC ID vpc-0df59c88041a47a21 DNS resolution Enabled Main network ACL acl-03f8359c1e958418e IPv6 CIDR (Network border group) - Encryption control ID -	State Available Tenancy default Default VPC No Network Address Usage metrics Disabled Encryption control mode -	Block Public Access Off DHCP option set dopt-0712af0b9a9f36d82 IPv4 CIDR 10.0.0.0/16 Route 53 Resolver DNS Firewall rule groups C	DNS hostnames Disabled Main route table rtb-0be5696b158135098 IPv6 pool - Owner ID 068371986937
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Resource map CIDRs Flow logs Tags Related resources Integrations

Resource map ⓘ

Show all details

VPC Subnets (0) Route tables (1)

NOTE : with the creation of VPC , a default route table and nacl will be created with it

Creation of subnet

Step 1 Choose the subnet option below the your VPC

Step 2 default subnet will be shown , to create new subnet choose create subnet

VPC dashboard <

Filter by VPC ▾

AWS Global View 

▼ **Virtual private cloud**

Your VPCs

[Subnets](#)

Subnets (3) [info](#)

Last updated 44 minutes ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

☐	Name ▾	Subnet ID ▾	State ▾	VPC ▾	Block Public... ▾	IPv4 CIDR
☐	default-1	subnet-01c384fb8135da73b	Available	vpc-02ba652b313394e79 def...	Off	172.31.0.0/20
☐	default-2	subnet-06909085a746057cf	Available	vpc-02ba652b313394e79 def...	Off	172.31.16.0/20
☐	default-3	subnet-01cb6bf9ca416c4ed	Available	vpc-02ba652b313394e79 def...	Off	172.31.32.0/20

Step 3 choose your new created VPC

Create subnet [info](#)

VPC

VPC ID
Create subnets in this VPC.

vpc-0df59c88041a47a21 (new-vpc) ▾

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Step 4 Then for making of the public subnet choose the following information

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

Public-subnet

The name can be up to 256 characters long.

Availability Zone [info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1a ▾

IPv4 VPC CIDR block [info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16 ▾

IPv4 subnet CIDR block

10.0.1.0/24 256 IPs

< > ^ v

▼ **Tags - optional**

Step 5 Then for making of the private subnet , choose the following information

You can add 49 more tags.

[Remove](#)

Subnet 2 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

Private-subnet

The name can be up to 256 characters long.

Availability Zone [info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1a ▾

IPv4 VPC CIDR block [info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16 ▾

IPv4 subnet CIDR block

10.0.2.0/24 256 IPs

< > ^ v

Step 6 choose create subnet

You can see the following information

VPC dashboard <

Filter by VPC ▾

AWS Global View ⓘ

▼ **Virtual private cloud**

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Subnets (5) [Info](#)

Last updated less than a minute ago [Actions](#) [Create subnet](#)

Q Find subnets by attribute or tag

☐	Name ▾	Subnet ID ▾	State ▾	VPC ▾	Block Public... ▾	IPv4 CIDR
☐	default-1	subnet-01c384fb8135da73b	Available	vpc-02ba652b313394e79 def...	Off	172.31.0.0/20
☐	Private-subnet	subnet-0909ab2f61f35dd66	Available	vpc-0df59c88041a47a21 new...	Off	10.0.2.0/24
☐	default-2	subnet-06909085a746057cf	Available	vpc-02ba652b313394e79 def...	Off	172.31.16.0/20
☐	default-3	subnet-01cb6bf9ca416c4ed	Available	vpc-02ba652b313394e79 def...	Off	172.31.32.0/20
☐	Public-subnet	subnet-0a3a2dff96c9a1e52	Available	vpc-0df59c88041a47a21 new...	Off	10.0.1.0/24

Creation of internet gateway

Step 1 choose the internet gateway below the route table

As you can ,there is a internet gateway for the default vpc now in order to create new internet gateway choose create internet gateway

VPC dashboard <

Filter by VPC ▾

AWS Global View ⓘ

▼ **Virtual private cloud**

Internet gateways (1) [Info](#)

[Actions](#) [Create internet gateway](#)

Q Find internet gateways by attribute or tag

☐	Name ▾	Internet gateway ID ▾	State ▾	VPC ID ▾	Owner
☐	default-ig	igw-0dc259dbd210a31c5	Attached	vpc-02ba652b313394e79 default-vpc	068371986937

Step 2 choose the following information

VPC > **Internet gateways** > Create Internet gateway

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

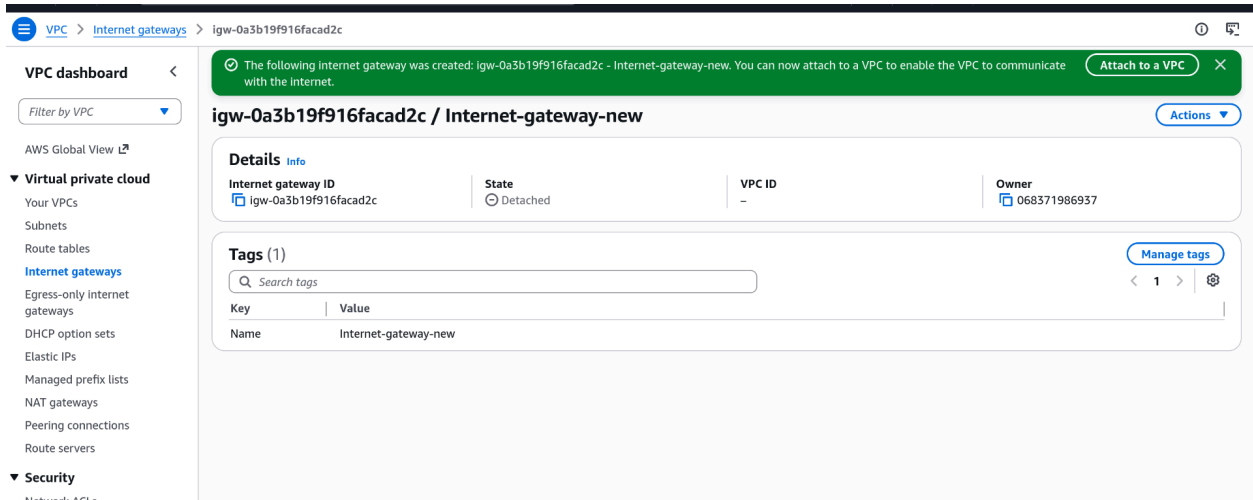
Key **Value - optional** [Remove](#)

[Add new tag](#)

You can add 49 more tags.

[Cancel](#) [Create Internet gateway](#)

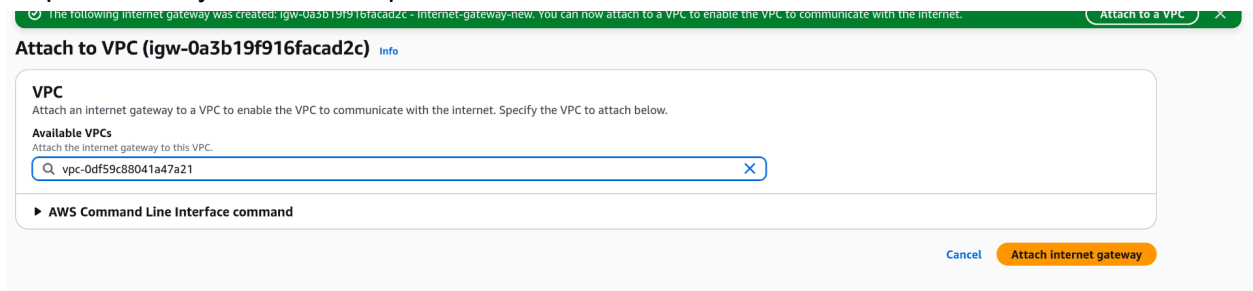
Step 3 choose create internet gateway ,and the following information will be created



Now it is not yet connected to vpc , so first we have to attached it to the VPC

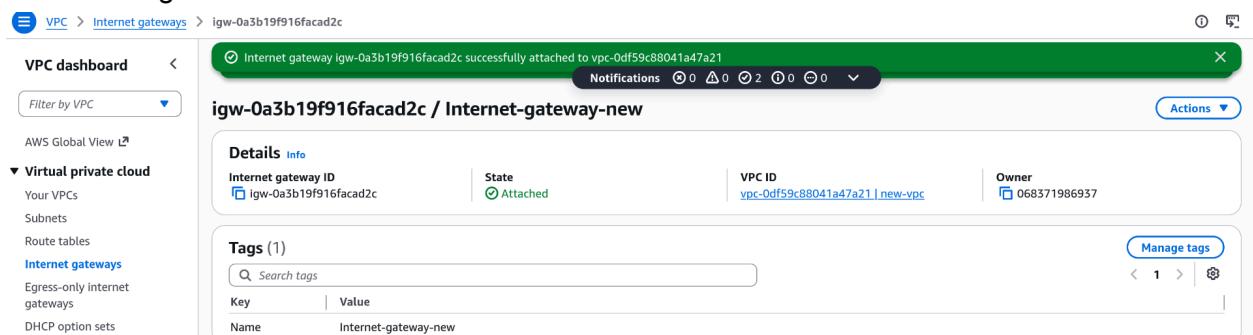
Step 4 choose attach to a Vpc as shown in green dialog box

Step 5 choose your created Vpc



Step 6 Attached Internet gateway

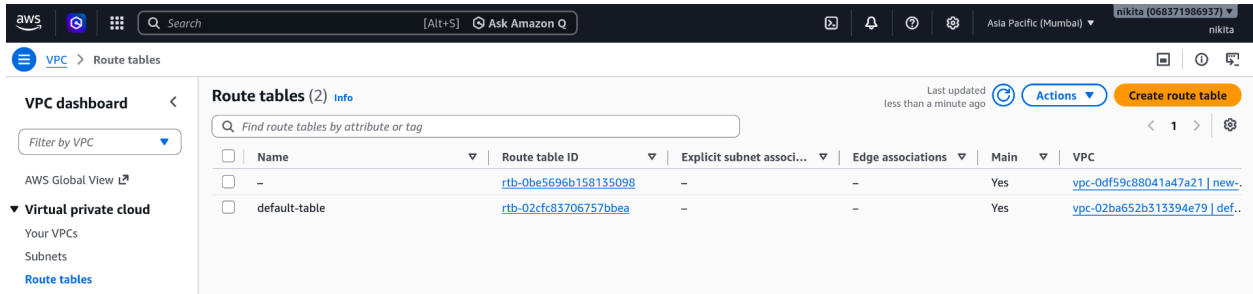
The following will be shown



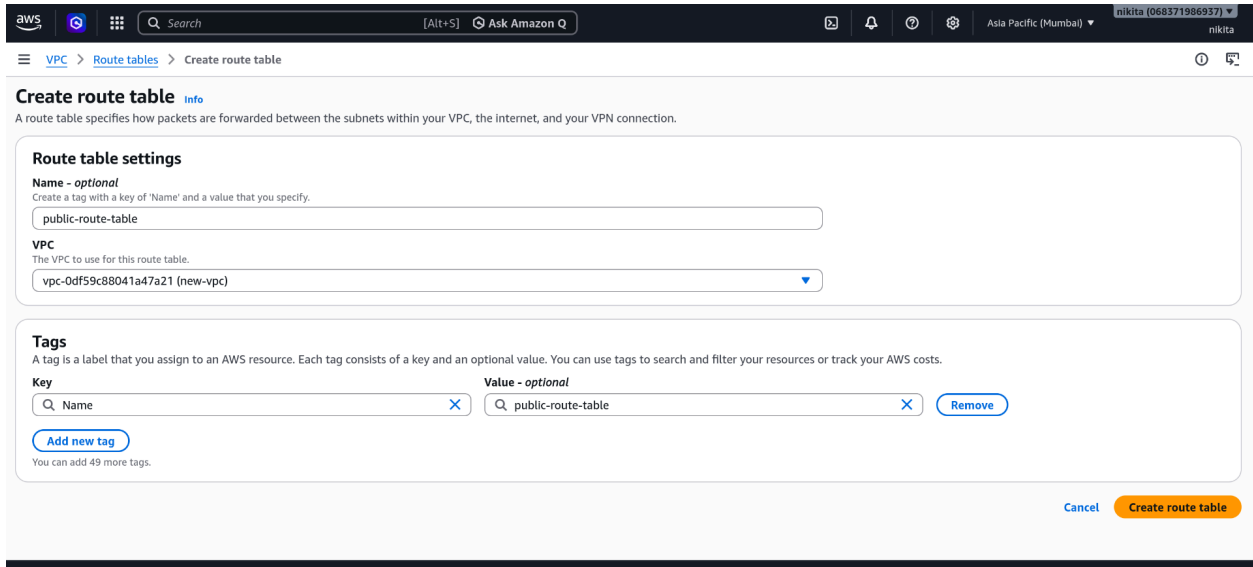
Creation of route tables

There will be two route tables, one that is default and other one will be created with your Vpc

Step 1 choose the route table option above the internet gateway , you can see the following in the route table dashboard

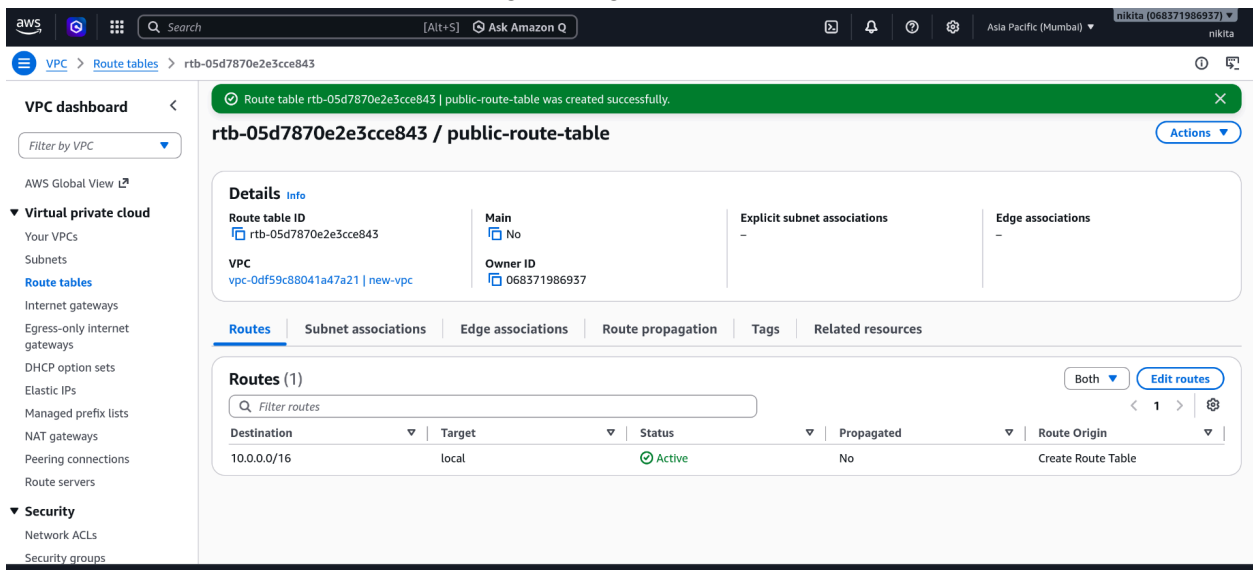


Step 2 choose create route table and it is for public choose the following information



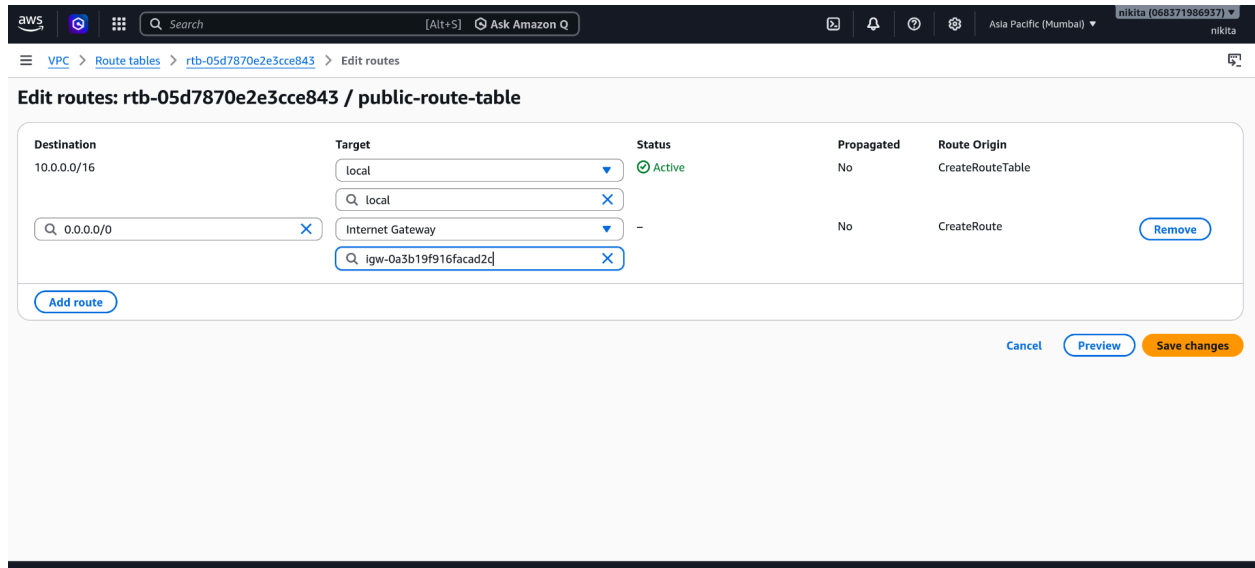
Step 3 create route table

In this we can see it has local , meaning it can go outside this



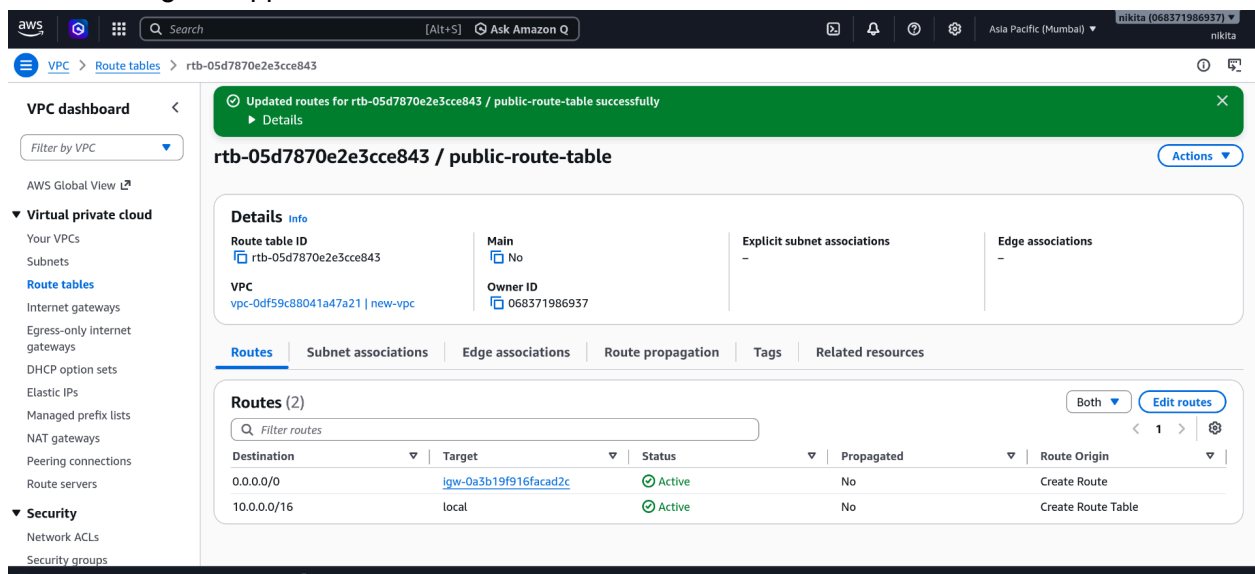
Step 4 So to make it public , selecting the route table , edit route , add route ,

Step 5 select 0.0.0.0/0 and then internet gateway , then choose the newly created internet gateway , that is attached to you newly created VPC



Step 5 click on save changes

The following will appear



Now do not edit anything in vpc created table just name it private-route , the following details can be seen

The screenshot shows the AWS VPC console with the 'Route tables' page. The left sidebar shows the 'VPC dashboard' with a search bar and a list of VPC resources. The main content area displays a table of route tables. The 'private-route' table is selected, and its details are shown below the table.

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒ private-route	rtb-0be5696b158135098	-	-	Yes	vpc-0df59c88041a47a21 new-
☐ default-table	rtb-02cfc83706757bbee	-	-	Yes	vpc-02ba652b313394e79 def..
☐ public-route-table	rtb-05d7870e2e3cce843	-	-	No	vpc-0df59c88041a47a21 new-

Below the table, the details for the selected 'private-route' table are shown, including its ID 'rtb-0be5696b158135098' and a link to 'private-route'.

Now we have to associate the subnet with the route-table

Step 6 Select the public-route-table

Step 7 select the subnet association

The screenshot shows the AWS VPC console with the 'Route tables' page. The 'public-route-table' is selected, and the 'Subnet associations' tab is active. The 'Explicit subnet associations' section shows no associations.

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
No subnet associations			

Below the table, a message states: 'You do not have any subnet associations.'

Step 8 edit subnet association , choose the public subnet and save association , how public subnet is associated with public route table

You have successfully updated subnet associations for rtb-0be5696b158135098 / private-route.

Route tables (1/3) Info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒ private-route	rtb-0be5696b158135098	subnet-0909ab2f61f35d...	-	Yes	vpc-0df59c88041a47a21 new-
☐ public-route-table	rtb-05d7870e2e3cce843	subnet-0a3a2df96c9a1e...	-	No	vpc-0df59c88041a47a21 new-
☐ default-table	rtb-02cf83706757bbea	-	-	Yes	vpc-02ba652b313394e79 def..

rtb-0be5696b158135098 / private-route

Details Routes Subnet associations Edge associations Route propagation Tags Related resources

Details

Route table ID rtb-0be5696b158135098 VPC vpc-0df59c88041a47a21 new-vpc	Main Yes Owner ID 068371986937	Explicit subnet associations subnet-0909ab2f61f35dd66 / Private-subnet	Edge associations -
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Creation of NACL

Step 1 select the network ACL in the security option

Network ACLs (2) Info

Find Network ACLs by attribute or tag

Name	Network ACL ID	Associated with	Default	VPC ID	Inbound
-	acl-03f8359c1e958418e	2 Subnets	Yes	vpc-0df59c88041a47a21 new-vpc	2 Inbound
default	acl-07791e08035555224	3 Subnets	Yes	vpc-02ba652b313394e79 default-vpc	2 Inbound

As you can see there are 2 nacl in this one is created with our newly ACL and second id the default one, give the newly created name "new-Nacl"

In this we have to create just one more NACL for our public one, so like on the create network ACL

Step 2 select the following information

Create network ACL Info

A network ACL is an optional layer of security that acts as a firewall for controlling traffic in and out of a subnet.

Network ACL settings

Name - optional
Creates a tag with a key of 'Name' and a value that you specify.

public-nacl

VPC
VPC to use for this network ACL.

vpc-0df59c88041a47a21 (new-vpc)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Name

Value - optional

public-nacl

Remove tag

Add tag

You can add 49 more tags

Cancel Create network ACL

Step 3 click on create network ACL the following information can be seen

Network ACLs (3) Info

Find Network ACLs by attribute or tag

☐	Name	Network ACL ID	Associated with	Default	VPC ID	Inbound
☐	new-nacl	acl-03f8359c1e958418e	2 Subnets	Yes	vpc-0df59c88041a47a21 / new-vpc	2 Inbound
☐	default	acl-07791e08035555224	3 Subnets	Yes	vpc-02ba652b313394e79 / default-vpc	2 Inbound
☐	public-nacl	acl-0c8e2cef537a4ebd0	-	No	vpc-0df59c88041a47a21 / new-vpc	1 Inbound

Select a network ACL

Step 4 select the public-nacl, click on subnet association

Network ACLs (1/3) Info

Find Network ACLs by attribute or tag

☒	Name	Network ACL ID	Associated with	Default	VPC ID	Inbound
☐	new-nacl	acl-03f8359c1e958418e	2 Subnets	Yes	vpc-0df59c88041a47a21 / new-vpc	2 Inbound
☐	default	acl-07791e08035555224	3 Subnets	Yes	vpc-02ba652b313394e79 / default-vpc	2 Inbound
☒	public-nacl	acl-0c8e2cef537a4ebd0	-	No	vpc-0df59c88041a47a21 / new-vpc	1 Inbound

acl-0c8e2cef537a4ebd0 / public-nacl

Details | Inbound rules | Outbound rules | **Subnet associations** | Tags

Subnet associations

Filter subnet associations

Name	Subnet ID	Associated with	Availability Zone	IPv4 CIDR	IPv6 CIDR
No subnets in this region are associated with this network ACL.					

Step 5 click on edit subnet association edit , choose the public subnet ,then save changes

aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) nikita (068371986937) nikita

VPC > Network ACLs > acl-0c8e2cef537a4ebd0 / public-nacl > Edit subnet associations

Edit subnet associations Info

Change which subnets are associated with this network ACL.

Available subnets (1/2)

Filter subnet associations

☐	Name	Subnet ID	Associated with	Availability Zone	IPv4 CIDR	IPv6 CIDR
☐	Private-subnet	subnet-0909ab2f61f35dd66	acl-03f8359c1e958418e / new-...	aps1-az1 (ap-south-1a)	10.0.2.0/24	-
☒	Public-subnet	subnet-0a3a2dff96c9a1e52	acl-03f8359c1e958418e / new-...	aps1-az1 (ap-south-1a)	10.0.1.0/24	-

Selected subnets

subnet-0a3a2dff96c9a1e52 / Public-subnet X

Cancel Save changes

You can see the following information

aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) nikita (068371986937) nikita

VPC > Network ACLs

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only Internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Security

- Network ACLs
- Security groups

You have successfully updated subnet associations for acl-0c8e2cef537a4ebd0 / public-nacl.

Network ACLs (1/3) Info

Find Network ACLs by attribute or tag

☐	Name	Network ACL ID	Associated with	Default	VPC ID	Inbound
☐	new-nacl	acl-03f8359c1e958418e	subnet-0909ab2f61f35dd66 / Private-subnet	Yes	vpc-0df59c88041a47a21 / new-vpc	2 Inbound
☐	default	acl-07791e08035555224	3 Subnets	Yes	vpc-02ba652b313394e79 / default-vpc	2 Inbound
☒	public-nacl	acl-0c8e2cef537a4ebd0	subnet-0a3a2dff96c9a1e52 / Public-subnet	No	vpc-0df59c88041a47a21 / new-vpc	1 Inbound

acl-0c8e2cef537a4ebd0 / public-nacl

Details Inbound rules Outbound rules Subnet associations Tags

Details

Network ACL ID acl-0c8e2cef537a4ebd0	Associated with subnet-0a3a2dff96c9a1e52 / Public-subnet	Default No	VPC ID vpc-0df59c88041a47a21 / new-vpc
Owner 068371986937			

Now blocking of the 100.100.100.100 ssh and ping for the public subnet

Step 6 select the public -nacl ,click on inbound rules

Step 7 edit inbound rules and choose the following information

Edit inbound rules [Info](#)
Inbound rules control the incoming traffic that's allowed to reach the VPC.

Rule number Info	Type Info	Protocol Info	Port range Info	Source Info	Allow/Deny Info	
100	SSH (22)	TCP (6)	22	192.168.31.13/24	Allow	Remove
90	SSH (22)	TCP (6)	22	100.100.100.100/32	Deny	Remove
91	All ICMP - IPv4	ICMP (1)	All	100.100.100.100/32	Deny	Remove
101	HTTP (80)	TCP (6)	80	0.0.0.0/0	Allow	Remove
110	HTTPS (443)	TCP (6)	443	0.0.0.0/0	Allow	Remove
120	All traffic	All	All	0.0.0.0/0	Allow	Remove
*	All traffic	All	All	0.0.0.0/0	Deny	

[Add new rule](#) [Sort by rule number](#)

[Cancel](#) [Preview changes](#) [Save changes](#)

Step 8 Now select the outbound rules and do the following configuration

Edit outbound rules [Info](#)
Outbound rules control the outgoing traffic that's allowed to leave the VPC.

Rule number Info	Type Info	Protocol Info	Port range Info	Destination Info	Allow/Deny Info	
	All traffic	All	All	0.0.0.0/0	Allow	Remove
*	All traffic	All	All	0.0.0.0/0	Deny	

[Add new rule](#) [Sort by rule number](#)

[Cancel](#) [Preview changes](#) [Save changes](#)

Creation of EC2

Step 1 Choose the ec2 option and launch instance and write the following information

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

 [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances [Info](#)

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-090d68841c2a28756

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

linux

[Create new key pair](#)

Do the following changes in the network settings

▼ Network settings [Info](#)

VPC - *required* | [Info](#)

vpc-0df59c88041a47a21 (new-vpc)
10.0.0.0/16



Subnet | [Info](#)

subnet-0a3a2dff96c9a1e52

Public-subnet

VPC: vpc-0df59c88041a47a21 Owner: 068371986937 Availability Zone: ap-south-1a (aps1-az1)
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24

[Create new subnet](#)

Auto-assign public IP | [Info](#)

Enable

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - *required*

launch-wizard-8

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and . _ - / () # , @ [] + = & ; ! \$ *

Description - *required* | [Info](#)

launch-wizard-8 created 2026-09-04T14:52:40.977Z

Inbound Security Group Rules

EC2 / INSTANCES / Launch an instance

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 49.36.185.10/32) Remove

Type	Protocol	Port range	Source type	Name	Description - optional
ssh	TCP	22	My IP	49.36.185.10/32	e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0) Remove

Type	Protocol	Port range	Source type	Source	Description - optional
HTTP	TCP	80	Anywhere	0.0.0.0/0	e.g. SSH for admin desktop

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Step 2 launch instance

Now to make the instance webserver ,select the webserver,and then click on connect

EC2 / INSTANCES / Connect to instance

Connect to Linux instance

Connect to an instance to test, troubleshoot, deploy code, update software etc. You can use browser based methods or an SSH client.

Instance info
 Instance: i-011aea115ca908605 (webserver)
 State: Running
 Status: 3/3 checks passed

VPC
 vpc-0df59c88041a47a21

Subnet
 subnet-0a3a2dff96c9a1e52

Security groups
 sg-0b77d4909fd419e4c (launch-wizard-9)

In web browser | In SSH client

Choose how to connect

☒ Public subnet access | Managed SSH keys
EC2 Instance Connect
 Use for instances with public IPs or accessible via standard routing. Requires network connectivity to your instance. Connect using temporary SSH keys managed by AWS.

☐ Private subnet access | Managed SSH keys
EC2 Instance Connect Endpoint
 Use for instances in private subnets. Traffic is routed through a VPC endpoint. Connect using temporary SSH keys managed by AWS.

☐ Maximum security | IAM authentication
SSM Session Manager
 Uses IAM-based authentication. Connect without SSH keys, bastion hosts, or opening any inbound ports.

☐ Direct | Troubleshoot
EC2 Serial Console
 Access your instance's serial port for troubleshooting boot, network, and OS issues. Works without network connectivity. Requires account-level enablement.

Settings

Choose public IP address

☒ IPv4 address (13.235.2.106) ☐ IPv6 address

Cancel Connect assist Connect

Step 3 Then run command

1.Sudo apt update

```
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-0-1-118:~$ sudo apt update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
```

2.Then sudo apt install apache2 -y

3.sudo systemctl enable --now apache2

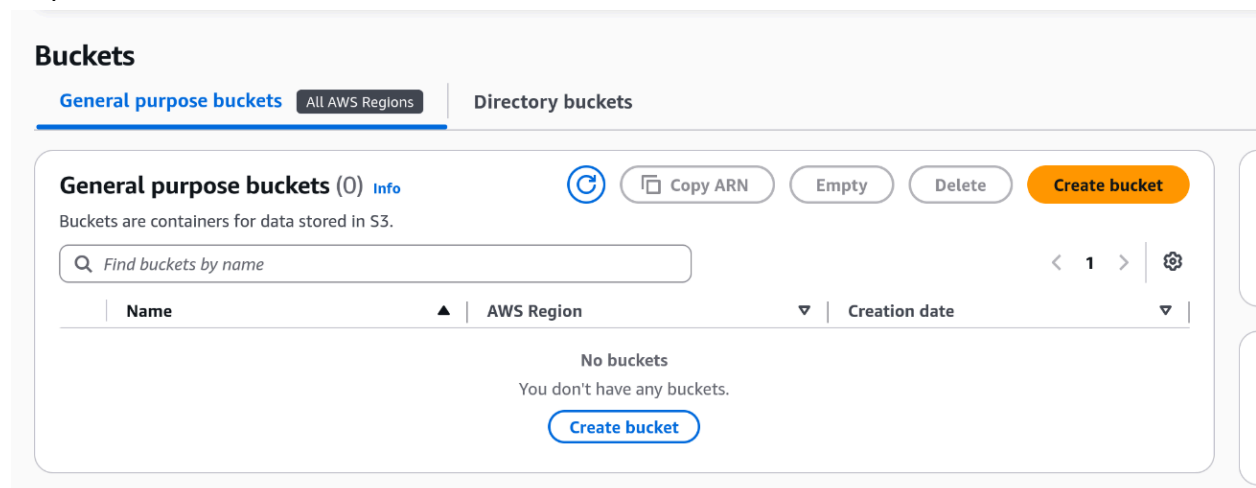
```
Try: sudo apt install <deb name>
ubuntu@ip-10-0-1-118:~$ sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
ubuntu@ip-10-0-1-118:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
```

4.Then sudo systemctl status apache2

```
ubuntu@ip-10-0-1-118:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-09-05 06:55:06 UTC; 1min 3s ago
     Invocation: 9ab55727321c4caf99b930a0ed1176a9
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 2740 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 55 (limit: 627)
    Memory: 5.8M (peak: 6M)
       CPU: 91ms
    CGroup: /system.slice/apache2.service
            └─2740 /usr/sbin/apache2 -k start -DFOREGROUND
            └─2756 /usr/sbin/apache2 -k start -DFOREGROUND
```

Creation of S3 bucket

Step 1 Select the s3 from the console



Step 2 select create bucket , the choose the following information

Amazon S3 > Buckets > Create bucket

Asia Pacific (Mumbai) ap-south-1

Bucket type

Info

☒ General purpose

Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ Directory

Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace

Choose the namespace where you want to create your bucket. [Learn more](#)

☒ Global namespace

By default, S3 creates general purpose buckets in the global namespace.

☐ Account Regional namespace (recommended)

General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name

Info

new-test-bucket-32

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

Object Ownership

Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

Amazon S3 > Buckets > Create bucket

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ Block all public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ Block public access to buckets and objects granted through new access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ Block public access to buckets and objects granted through any access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

☐ Block public access to buckets and objects granted through new public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ Block public and cross-account access to buckets and objects through any public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Turning off block all public access might result in this bucket and the objects within becoming public

AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☐ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can

Now we have to upload the images

Amazon S3 > Buckets > new-test-bucket-32

Successfully created bucket "new-test-bucket-32"

new-test-bucket-32

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems

Access Points

Objects (0)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

Name

Type

Last modified

Size

Storage class

No objects

You don't have any objects in this bucket.

Upload

Step 3 Click on upload

Amazon S3 > Buckets > new-test-bucket-32 > Upload

Upload [Info](#)

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (0)
All files and folders in this table will be uploaded.

Name

Folder

Type

Size

No files or folders
You have not chosen any files or folders to upload.

Destination [Info](#)

Destination
[s3://new-test-bucket-32](#)

► Destination details
Bucket settings that govern how objects stored in the specified destination

,the following will appear , after you have uploaded the following will appear

Amazon S3

Upload succeeded
For more information, see the Files and folders table.
After you navigate away from this page, the following information is no longer available.

Summary

Destination
[s3://new-test-bucket-32](#)

Succeeded
2 files, 292.7 KB (100.00%)

Failed
0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (2 total, 292.7 KB)

Name	Folder	Type	Size	Status	Error
pineapple.jpg	-	image/jpeg	291.6 KB	Succeeded	-
fruit.html	-	text/html	1.1 KB	Succeeded	-

Now to access the objects we can make policies and make it access publicly

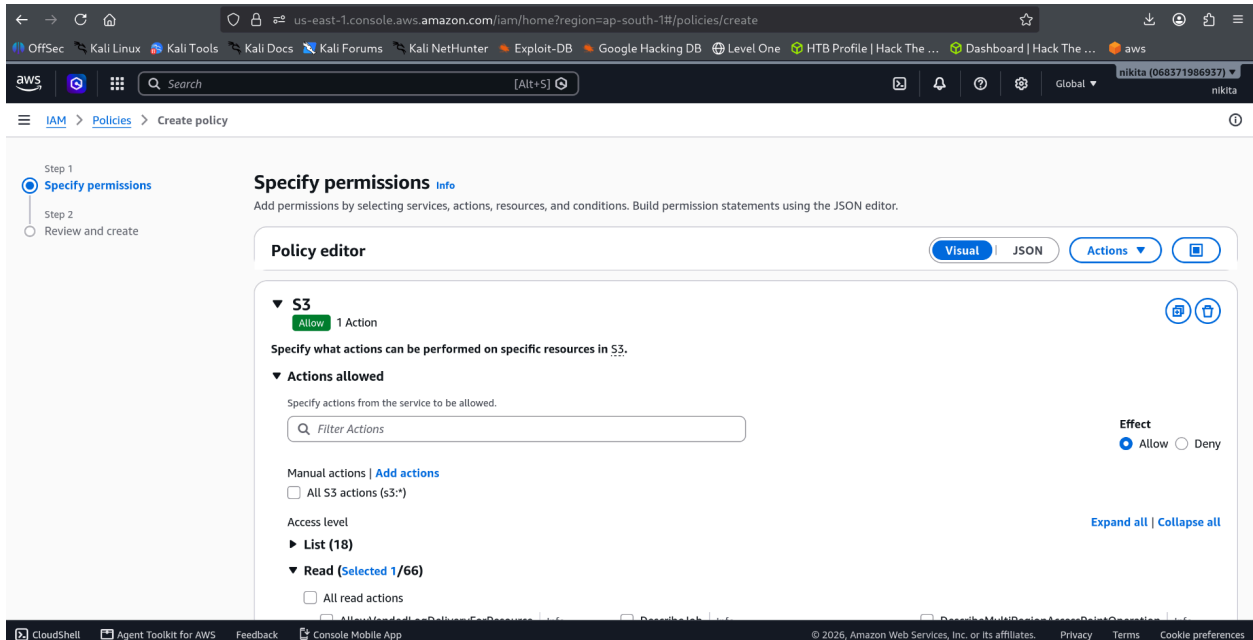
Step 1 go to the policy

Step 2 Create policy

Step 3 select the S3 service

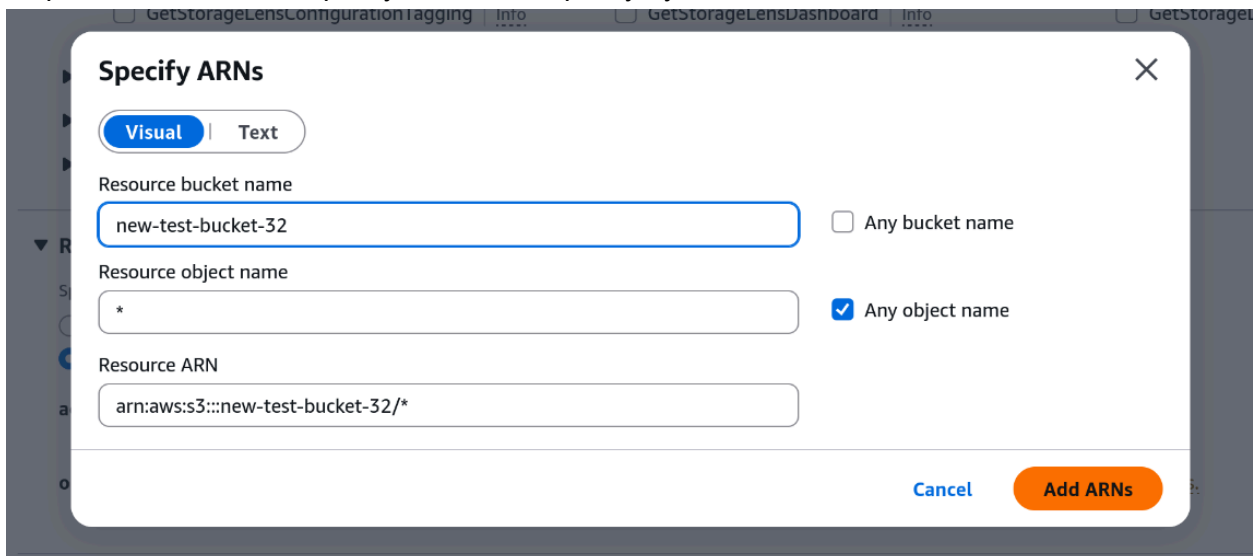
Step 4 select the Getobject

Step 5 now have to give the ARN to the specific



Step 6 Now on the object delete ADD ARN ,you can get the ARN ,when you select the object that you it to associate it with and go to the permissions

Step 7 select the bucket policy ,edit bucket policy , you can the bucket ARN

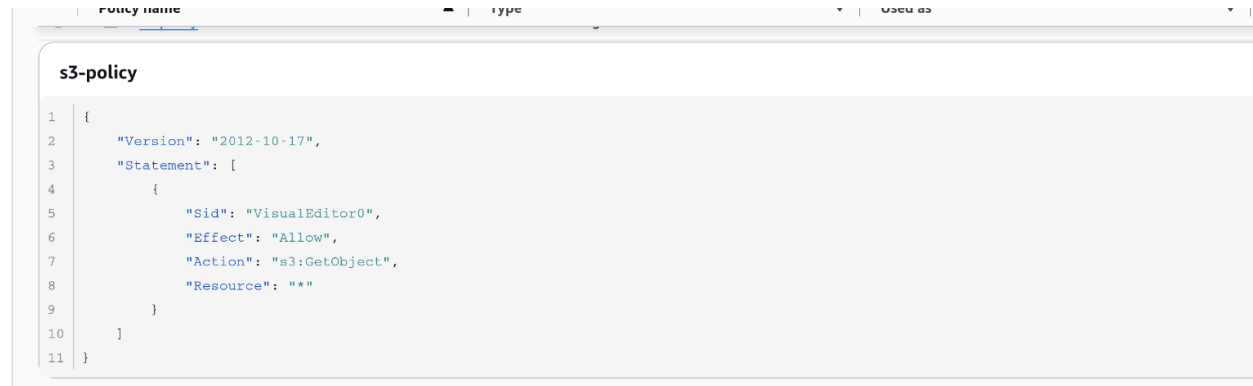


Step 8 next , you can follow the same steps with other object

Now have to associate the policies with the object of the bucket

Step 1 in the policy ,select the custom managed

Step 2 click on the plus sign,and the json will appear



The screenshot shows the AWS IAM console interface for a custom policy named "s3-policy". The policy is of type "Custom managed". Below the policy name, there is a JSON editor showing the following statement:

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "VisualEditor0",
6       "Effect": "Allow",
7       "Action": "s3:GetObject",
8       "Resource": "*"
9     }
10  ]
11 }
```

Step 3 select the object ,then permissions

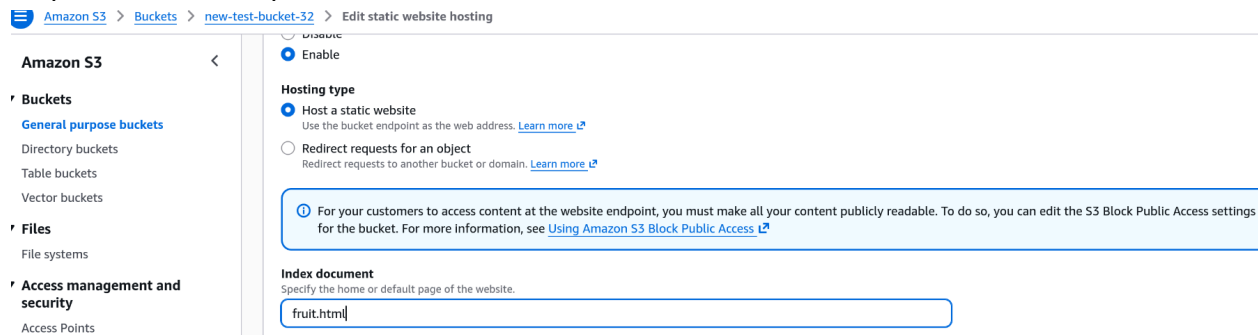
Step 4 Paste the json in the bucket policy and the principal="*"

Step 5 Then you can access the object from your webserver

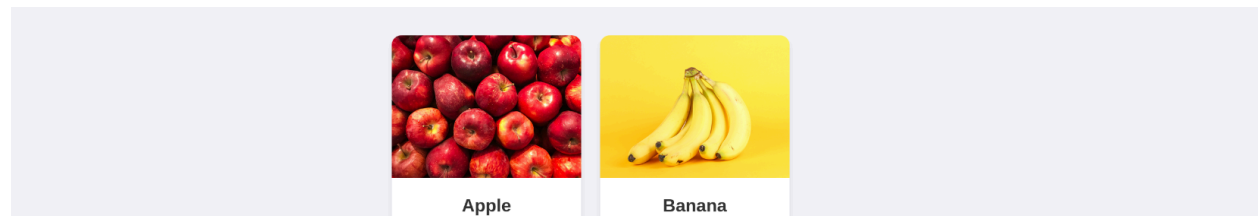


You can also access it by enabling of the static website

- Step 1 select the object
- Step 2 select properties
- Step 3 enable static website
- Step 4 follow the steps

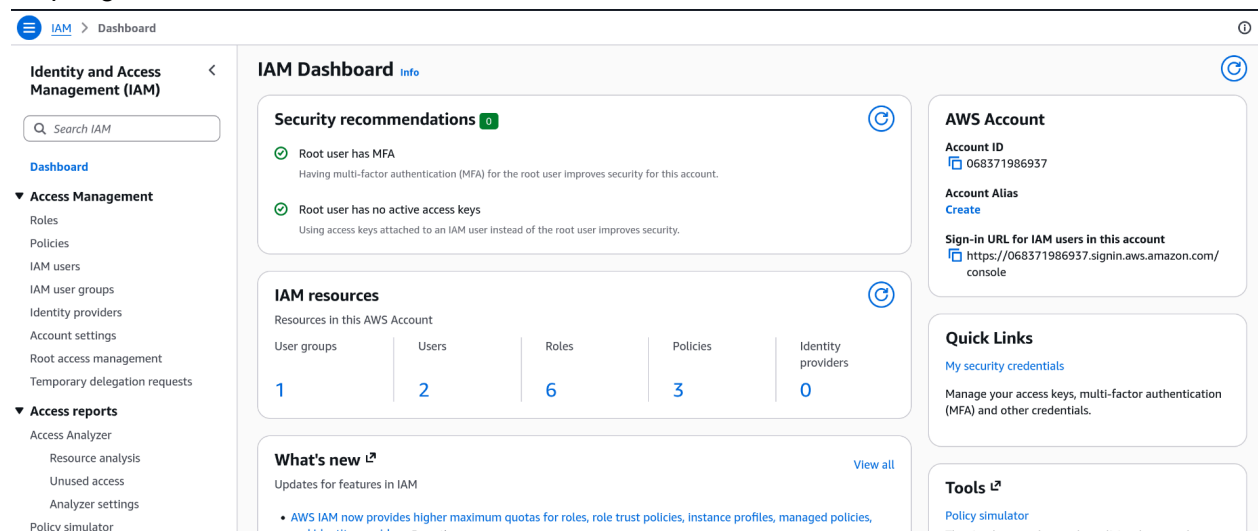


Now you can access it by , clicking on the link given in the static website



Creation of IAM Roles

Step 1 go to the IAM dashboard



Step 2 select roles ,then create role

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

- Roles
- Policies
- IAM users
- IAM user groups
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Access reports

- Access Analyzer
 - Resource analysis
 - Unused access
 - Analyzer settings
- Policy simulator

Role deleted ec2-role.

Roles (5) Info

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Search

☐	Role name	Trusted entities	Last activity
☐	AWSServiceRoleForAutoScaling	AWS Service: autoscaling (Service-Linked Role)	18 days ago
☐	AWSServiceRoleForElasticLoadBalancing	AWS Service: elasticloadbalancing (Service-Linked Role)	18 days ago
☐	AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2 (Service-Linked Role)	14 minutes ago
☐	AWSServiceRoleForSupport	AWS Service: support (Service-Linked Role)	-
☐	AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service-Linked Role)	-

Roles Anywhere Info

Authenticate your non AWS workloads and securely provide access to AWS services.

Access AWS from your non AWS workloads X.509 Standard Temporary credentials

Step 3 select the following

IAM > Roles > Create role

Step 1: Select trusted entity

Step 2: Add permissions

Step 3: Name, review, and create

Select trusted entity Info

Trusted entity type

- ☒ AWS service
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ Web Identity
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ SAML 2.0 federation
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EC2

Choose a use case for the specified service.

Use case

EC2

Step 4 choose the following policies

Search IAM

Dashboard

Access Management

- Roles
- Policies
- IAM users
- IAM user groups
- Identity providers
- Account settings

Permissions policies (1) Info

You can attach up to 20 managed policies.

Filter by Type

Search

All types

☐	Policy name	Type	Attached entities
☐	AmazonS3ReadOnlyAccess	AWS managed	1

Now we have to attach the bucket with the EC2
Now go to EC2 ,select the ec2 that you have created , click actions
Step1 choose security— modify IAM role
Step 2 choose the created roles

Step 3 update IAM role, and it will be done

Creation of elastic beanstalk

Go to the dashboard of the elastic beanstalk

Elastic Beanstalk

Applications

Environments

Change history

Compute

AWS Elastic Beanstalk

End-to-end web application management.

AWS Elastic Beanstalk is an easy-to-use service for deploying and scaling web applications and services developed with Java, .NET, PHP, Node.js, Python, Ruby, Go, and Docker on familiar servers such as Apache, Nginx, Passenger, and IIS.

Get started

Easily deploy your web application in minutes.

Create application

Pricing

There's no additional charge for Elastic Beanstalk. You pay for Amazon Web Services resources that we create to store and run your web application, like Amazon S3 buckets and Amazon EC2 instances.

Getting started

Get started

You simply upload your code and Elastic Beanstalk automatically handles the deployment, from capacity provisioning, load balancing, and automatic scaling to web application health monitoring, with ongoing fully managed patch and security updates.
[Learn more](#)

CloudShell

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Step 1 click on the create application

Step 2 then choose the following information

Elastic Beanstalk

Applications

Environments

Change history

Elastic Beanstalk > Create environment

Environment details

Specify the primary details for this environment.

Basic details

Choose an application and name your environment.

Application name

Info

Select an existing application or create a new one.

Q mywebtest

X

Maximum of 100 characters.

A new application will be created

Domain name prefix - optional

Info

Optionally specify a unique domain name prefix. A default will be provided.

Leave blank for autogenerated value

Must be 4-63 characters. Letters, numbers, and hyphens are allowed. Cannot start or end with a hyphen.

Platform details

Info

Provide platform information that Elastic Beanstalk needs to create an environment.

Platform

Platform branch

Platform version

Select a platform

Select a branch

Select a version

Environment name - optional

Info

Optionally specify a unique environment name. A default will be provided.

Mywebtest-env

Must be 4-40 characters. Letters, numbers, and hyphens are allowed. Cannot start or end with a hyphen.

Environment description - optional

Info

Enter additional details for this environment to describe its attributes, purpose, etc.

Environment details

Maximum 200 characters.

CloudShell

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Elastic Beanstalk > Create environment

A new application will be created

Domain name prefix - optional [Info](#)
Optionally specify a unique domain name prefix. A default will be provided.

Must be 4-63 characters. Letters, numbers, and hyphens are allowed. Cannot start or end with a hyphen.

Environment description - optional [Info](#)
Enter additional details for this environment to describe its attributes, purpose, etc.

Maximum 200 characters.

Platform details [Info](#)
Provide platform information that Elastic Beanstalk needs to create an environment.

Platform **Platform branch** **Platform version**

Application code [Info](#)
Choose how to deploy this application.

☒ **Sample app**
Hello world!

☐ **Existing version**
No versions available for this application.

☐ **Local file**
Upload a file from your local machine.

☐ **S3 URI**
S3 object containing your application code.

Environment properties [Info](#)
Pass environment variables to your application as key-value pairs. Platform default variables can only have their values changed. They cannot be renamed or removed.

☐ Add environment properties

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Step 3 select the following infrastructure

Elastic Beanstalk > Create environment

Infrastructure - optional
Configure the compute capacity of your environment and Auto Scaling settings to optimize the number of instances used.

Environment tier [Info](#)
AWS Elastic Beanstalk has two types of environment tiers to support different types of web applications.

☒ **Web server environment**
Run a website, web application, or web API that serves HTTP requests.

☐ **Worker environment**
Run a worker application that processes long-running workloads on demand or performs tasks on a schedule.

Scaling
Configure Auto Scaling and load balancing settings for your environment.

Environment type [Info](#)
Select a single-instance or load-balanced environment.

☒ **Single instance**
A single-instance environment contains one Amazon EC2 instance with an Elastic IP address.

☐ **Load balanced**
A load-balanced and scalable environment uses the Elastic Load Balancing and Amazon EC2 Auto Scaling services to provision the Amazon EC2 instances that are required for your deployed application.

Min instances [Info](#)
The minimum number of EC2 instances for your environment.

1-10,000 instances

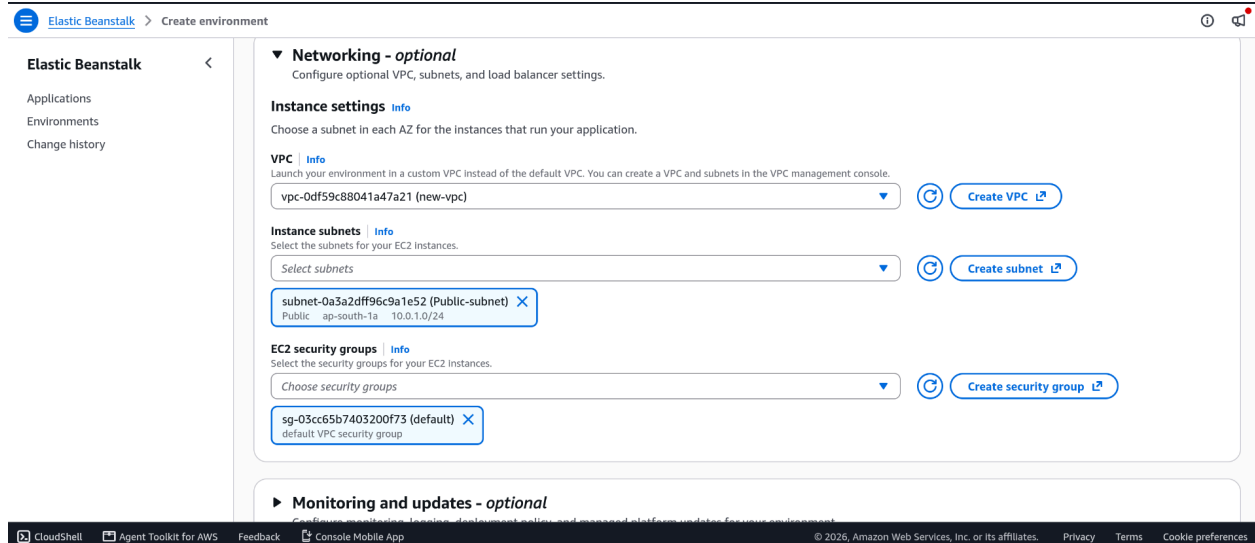
Max instances [Info
The maximum number of EC2 instances for your environment.

1-10,000 instances](#)

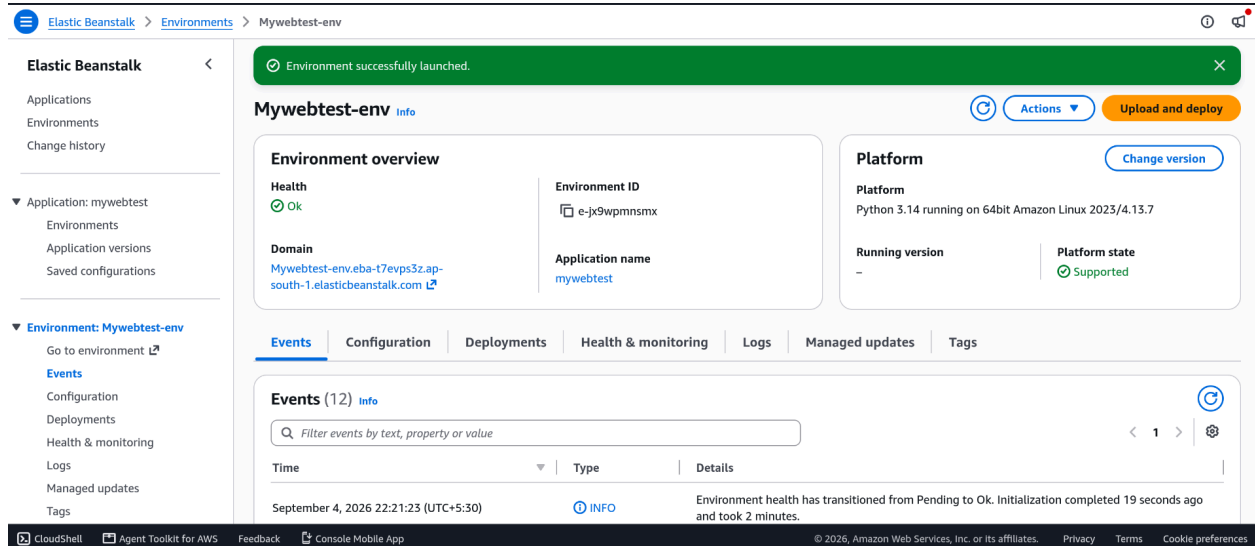
Auto Scaling configuration

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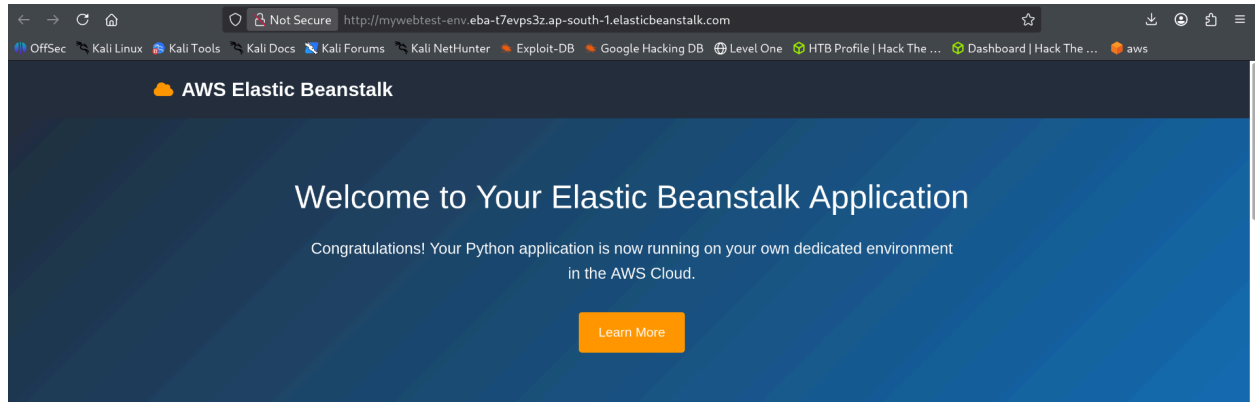
Step 4 select the following in the network



The following will tell you that environment is created



Step 5 Now you can open your website with the URL and you can see by clicking on it and following will appear



Benefits of AWS Elastic Beanstalk

Discover why thousands of developers rely on AWS Elastic Beanstalk to deploy and manage their applications.



EC2 to S3 access verification

Step 1 check whether the aws is installed or not run command : aws --version

```
ubuntu@ip-10-0-1-118:~$ aws --version
Command 'aws' not found, but can be installed with:
sudo apt install awscli
ubuntu@ip-10-0-1-118:~$
```

Step 2 install the AWScli

```
ubuntu@ip-10-0-1-118:~$ sudo apt update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
158 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-10-0-1-118:~$ sudo apt install awscli -y
Installing:
  awscli

Installing dependencies:
  docutils-common  libimagequant0  liblcms2-2  libpaper2  libwebp7
```

Step 3 To check the details

```
Last login: Sat Sep  5 06:51:23 2026 from 13.233.177.4
ubuntu@ip-10-0-1-118:~$ aws sts get-caller-identity
{
  "UserId": "AROAQ722JXH4U5APW7WQA:i-011aea115ca908605",
  "Account": "068371986937",
  "Arn": "arn:aws:sts::068371986937:assumed-role/ec2-role/i-011aea115ca908605"
}
ubuntu@ip-10-0-1-118:~$
```

i-011aea115ca908605 (webserver)

PublicIPs: 13.233.231.24 PrivateIPs: 10.0.1.118

Now to test S3 access on this

Step 1 run command `aws s3 ls s3://new-test-bucket-32`

```
2026-09-04 16:49:29 elasticbeanstalk-ap-south-1-068371986937
2026-09-05 07:17:10 new-test-bucket-32
ubuntu@ip-10-0-1-118:~$
```

And this tells that the connection is successful